

Saad Alif A Zaman

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RESEARCH INTERESTS

• Computer Vision for Industrial & Infrastructure Inspection • Deep Learning under Data Scarcity • Computer Vision for Civil & Infrastructure Engineering • Structural Health Monitoring (SHM) and Non-Destructive Evaluation (NDE) for marine and civil infrastructure • Toward robotic and autonomous platforms for field inspection and environmental monitoring

RESEARCH EXPERIENCE

➤ **Research Assistant** (August 2025 – Present)

“[Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection](#)”

Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept. of Naval Architecture and Marine Engineering, BUET
Samiul Based Shuvo, Assistant Professor, Dept. of Biomedical Engineering, BUET

- Addressing a core bottleneck in applied computer vision, the lack of large labeled datasets in real world industrial settings, we proposed a k–n Fold Augmentation Cross Validation (k–n Fold ACV) protocol, together with a novel Interaction Effect (IE) Metric that quantifies order sensitivity among data augmentations. The framework was validated on a newly curated [NDBD](#) dataset of 35 high resolution ship hull images containing diverse biofouling patterns with 92 annotated instances. Eleven domain specific augmentations and 110 pairwise combinations were analyzed using YOLOv8m-seg as the base model, with cross-validation across YOLO11m-seg, and Mask R-CNN. The [code](#) and [dataset](#) are publicly available.

➤ **Undergraduate Thesis** Final Grade: A+ (4.00/4.00) (June 2023 – July 2024)

“[Developing a biofouling & corrosion identification and distribution system using video images based on deep learning](#)”

Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept of Naval Architecture and Marine Engineering, BUET

- Collected and curated a biofouling & corrosion dataset from Narayanganj Dockyard, Bangladesh and trained YOLOv8 segmentation variants (transfer learning) to detect and localize biofouling and steel-corrosion defects using video/images (fouling F1 up to 0.90); implemented annotation (CVAT → YOLO), cloud training (Colab/Kaggle), and produced a real-time detection prototype for automated hull inspection, reducing manual labor and enabling sustainable maritime maintenance by minimizing fuel consumption and emissions.

➤ **Junior Year Project** Final Grade: A+ (4.00/4.00) (May 2022 – May 2023)

“[Designing an Inland 100 Passenger Ship for Inland waterways of Bangladesh](#)”

Supervisor: Dr. Md. Mashiur Rahaman, Professor, Dept of Naval Architecture and Marine Engineering, BUET

- Designed a conceptual 100-passenger inland vessel with an electric propulsion system tailored to Bangladesh’s inland waterway transport needs to reduce CO₂ emissions and improve energy efficiency compared to conventional diesel systems.

PUBLICATIONS

- Minhazul Islam, **Saad Alif-A Zaman**, Md. Mashiur Rahaman, “[Concept of an Electric-Powered Passenger Vessel in Bangladesh](#),” Journal of Maritime Research, vol. 20, no. 3, pp. 68–71, Dec. 2023
- Saad Alif-A Zaman, Samiul Based Shuvo, Kazi Naimul Hoque, "Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection," Preprint; in revision following peer review, targeting resubmission, 2026. [[SSRN link](#)]

EDUCATION

B.Sc. in Naval Architecture And Marine Engineering ([NAME](#)) (April 2019 – July 2024)

Bangladesh University of Engineering and Technology ([BUET](#))

Cumulative GPA: 3.37 / 4.00

Relevant Coursework: Marine Maintenance and Repair (corrosion, fatigue, and biofouling), Welding Technology (non-destructive testing, defect inspection), Finite Element Methods, Numerical Computations, Computer Programming (C/C++, FORTRAN), Applied Statistics

RESEARCH GRANTS, RECOGNITION & SCHOLARSHIPS

- RISE Student Research Grant 2023:**
Awarded a competitive Student Research Grant (BDT 65,790) by Research and Innovation center for Science and Engineering ([RISE](#)) for the research project: "Developing a biofouling & corrosion identification and distribution system using video images based on deep learning."

- **SPARRSO Research Fellowship Program 2025–26:**
Shortlisted (final round) among national applicants for a proposal-stage research fellowship with the Space Research and Remote Sensing Organization ([SPARRSO](#)), Bangladesh; invited to present the written proposal "*Development of a Fixed-Wing VTOL Quad Plane for Flood Monitoring & Disaster Response*" (design not yet implemented).
- **University Stipend Scholarship-2019, 2020, 2021 – BUET:**
Competitive merit-based financial award granted to high-performing undergraduate students based on academic results.
- **Rajshahi Board Scholarship HSC 2018:**
Merit-based national scholarship awarded for outstanding academic performance in the Higher Secondary Certificate (Grade 12 equivalent) examination.

STANDARDIZED TEST SCORES

International English Language Testing System (IELTS) Academic

(15th December, 2025)

Overall Band Score: 7.5 (Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 7.0)

WORK EXPERIENCE & INTERNSHIP

• Materials Development Instructor

Udvash Academic & Admission Care | Dhaka, Bangladesh

(April 2025 – August 2025)

Developed and reviewed assessment materials for SSC, HSC, engineering, and university admission examinations. Evaluated question quality, solution accuracy, and curriculum alignment. Contributed to model test preparation and academic content development.

• Academic Mentor (Mathematics & Physics)

(2019 – Present)

Provided one-to-one and small batch instruction to secondary and higher secondary students (Bangla and English medium curricula), from foundational concepts through advanced problem solving. Multiple students subsequently admitted to BUET, KUET, DU, and IUT.

• Industrial Internship – Shipyard Practice

Khulna Shipyard Limited | Khulna, Bangladesh

(October 2022 – November 2022)

TECHNICAL SKILLS

- **Programming:** Python, C, C++
- **Deep Learning:** Ultralytics YOLO, Mask R-CNN, PyTorch, TensorFlow
- **Data Annotation & Processing:** CVAT, Roboflow, LabelImg
- **Research Computing & Collaboration:** Google Colab, Kaggle

- **CAD software:** AutoCAD, Rhino
- **Finite Element Analysis software:** Abaqus
- **Document Preparation:** LaTeX, Prism, Microsoft Office

LEADERSHIP & VOLUNTEERING

• Founder, Monthly Reading Initiative (Telegram):

(March 2024 – Present)

Founded a monthly book and magazine reading platform on Telegram, which continues to be maintained by junior students and now includes regular quizzes and discussion activities.

• Badhon BUET Zone, Volunteer Member:

(May 2019 – July, 2024, Dhaka)

Active volunteer in a student-led blood donation organization; donated blood multiple times and participated in donation drives and awareness activities.

REFERENCES

Dr. Kazi Naimul Hoque

Associate Professor, RA and Thesis Supervisor
Dept. of Naval Architecture and Marine Engineering,
BUET, Dhaka-1000, Bangladesh

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[Faculty Profile](#)

Samiul Based Shuvo

Assistant Professor, RA Co-Supervisor
Dept. of Biomedical Engineering
BUET, Dhaka 1205, Bangladesh

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[Faculty Profile](#)

Dr. Md. Mashud Karim

Professor, Academic Advisor
Dept. of Naval Architecture and Marine Engineering,
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[Faculty Profile](#)